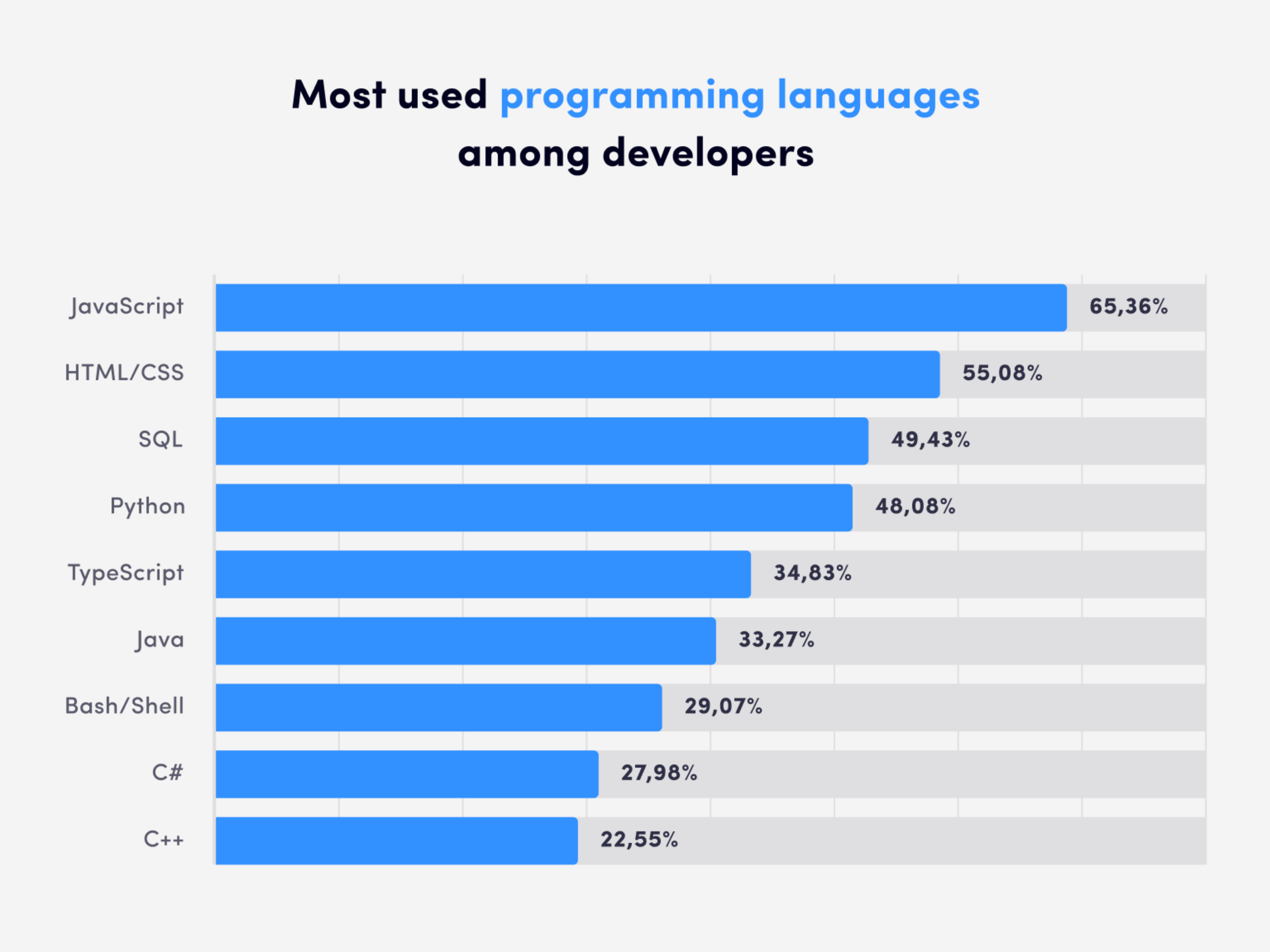
Javascript Fundamentals

What is best programming language ?

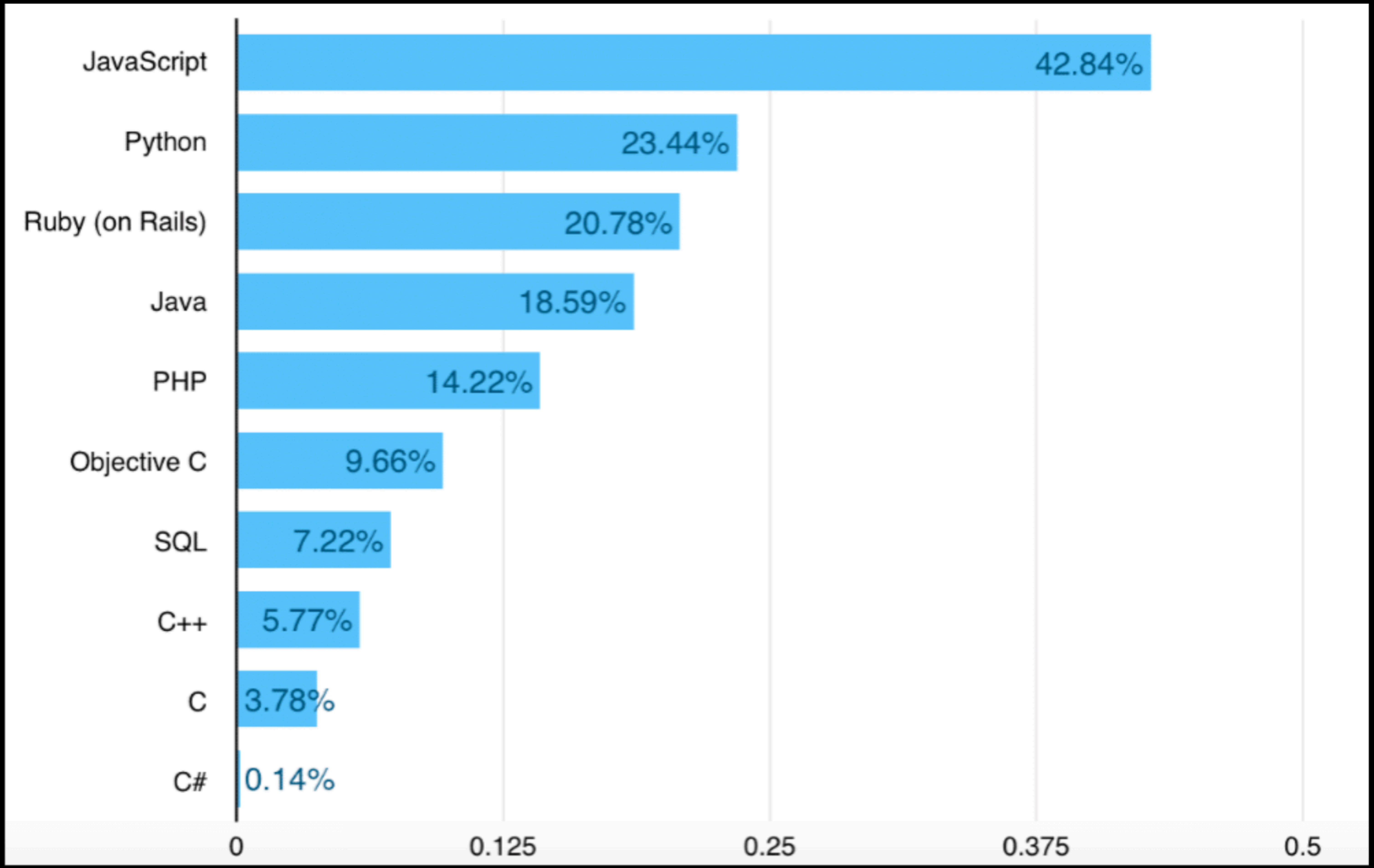
For beginner

The market



Most used programming languages
among developers
JavaScript
HTML/CSS
SQL
Python
TypeScript
Java
Bash/Shell
C#
C++
65,36%
55,08%
49,43%
48,08%
34,83%
33,27%
29,07%
27,98%
22,55%

The fastest learning language



The right way to learn code

Mastering one first

Everything else become easier

The right way to learn code

Clean Code

Advanced topics

Design Principle

Structure

FrameWork

Programming Language

Domain Knowledge

Learning source

- A-Z: MDN - Mozilla Developer Network
- Blog: Toidicodedao, The Notorious Snacky(fullsnack) - huytd
- Advanced: How javascript work - session stack
- Books
  - Clean Code
  - Pragmatic Programmer

Javascript Course

Basics

Functions

Module

Engine

Nodejs

Basics

Pass by value	Reference Type

Primitives	Operator	If else
Function	Array	Object

Primitives

Number	String	Boolean	Null
Undefined	Symbol	BigInt	

Primitives

When you declare a variable and assign it a primitive value (for example: let myName= Quylaokame), the computer allocates a memory space for the variable myName and directly stores the value Quylaokame in that memory space.

```
// you can not change value of a primitive
let myName = "Quylaokame";

let shortName = myName.slice(6);

console.log(shortName) // "kame";

console.log(myName) // "Quylaokame";

// you just can assign your variable to another value
myName = "kame";
```

String and Number

Computers do not store real numbers (such as 0.1, 3.14) exactly as we writ
them in the decimal system. Instead, they use the binary system and
a standard called IEEE 754 to represent these numbers.

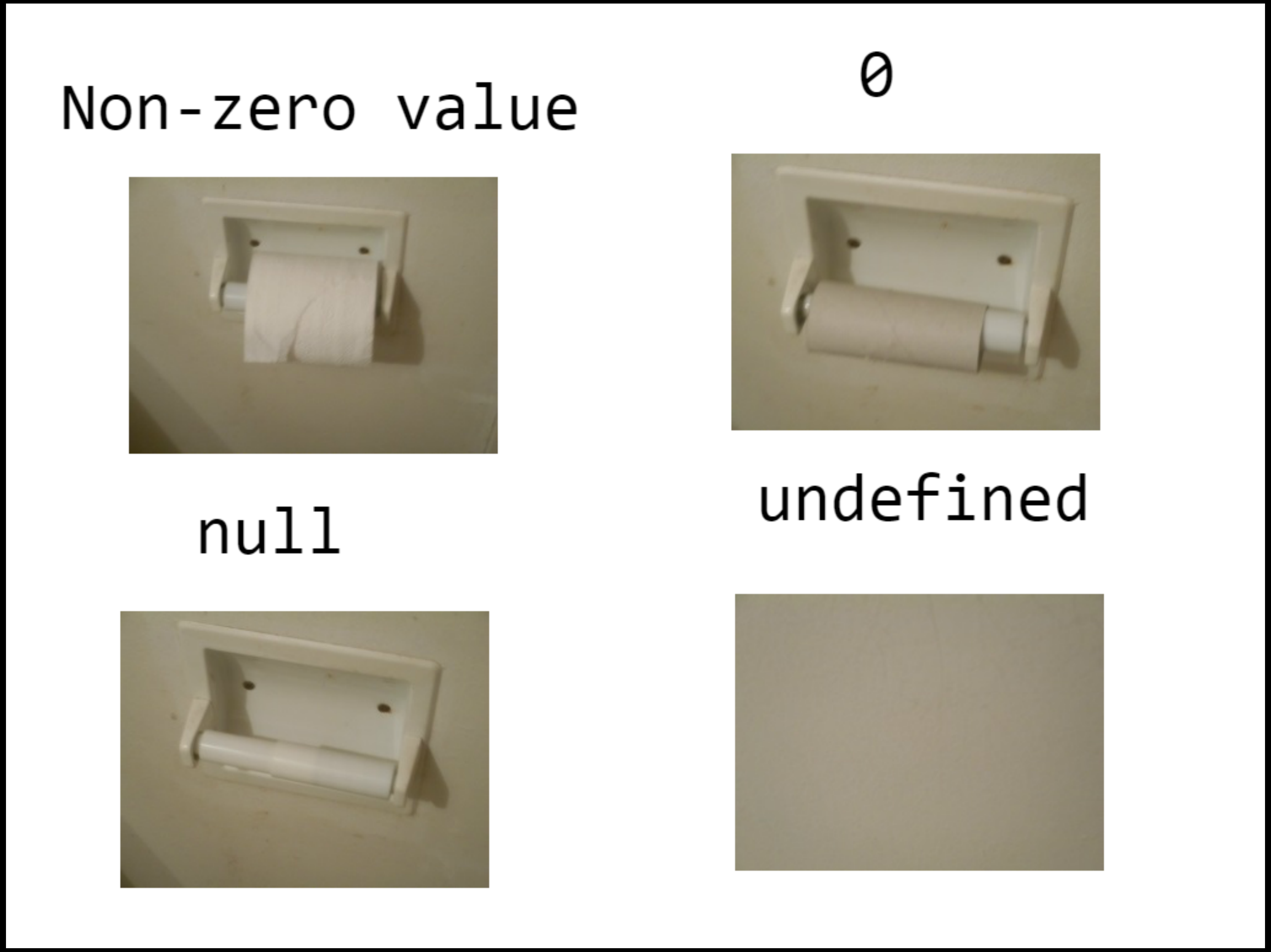
```
let x = 5 + 7;
// 12
x = 5 + "7";
// 57
x = 5 - "x";
// NaN
x = null - 15;
// -15
x = undefined - 15;
// NaN
```

```
typeof NaN// number

// careful with js float
x = 0.1 + 0.2;
(x === 0.3) // false

// converted string to number
let string = “-012.45”;
let score = "1.7e19"
num = Number(string);
let num = +string;
```

undefined

```javascript
let x;
// x is undefined
console.log(y);
//Uncaught ReferenceError: y is not defined

function add(x, y) {
    console.log(x, y)
    return x + y;
};
add(3);
// x = 3, y = undefined
typeof undefined; // "undefined";
```

null



Falsy values javascript

```
// falsy value javascript

Boolean(null); // false

!!undefined // false

!!0 // false
```

```
// falsy value javascript

!!"" // false

!!"0" // true

!!NaN // false
```

if else

```
x = 0; // falsy values…
if (x) // false

// the most difficult bug in js
if (x = 10) {
    // it is true
}
if (x = 0) {

} // it is false
```

```
x = 10;

switch (x) {

  case "10": console.log(“Hello”);
  // don't log anything

  case 10: alert("Hello");
  // it works
```

Array access (reference type)

```
var cars = ["Mazda", "Volvo", "BMW"];
var user = ["John", "Jim", "Jay"];

cars[0] // "Mazda";
cars[1] = "Toyota";

const [car1, car2, car3] = cars;
//car1: "Mazda", car2: “Toyota"
```

```
let arr = [0, 1, 2];
arr[4] = “4";
// [0, 1, 2, empty, 4]
console.log(arr[3]) // undefined

arr.length = 10;
// [0, 1, 2, empty, "new", empty × 5]
arr.length = 2;
console.log(arr) // [0, 1]
```

Array add, remove item

```
let letters = ["A","B","C"];
// add items to end of array
letters.push("D"); // 3
// return the length of fruits

// add items to start of array
letters.unshift("O"); // 4

// Remove an item from the end
let last = letters.pop(); // "D"

// Remove an item from beginning
let first = letters.shift(); // "O"
```

```
//Remove an item by index position
let secondItem = fruits.splice(1, 1);

delete letters[2];
console.log(letters);//  ["A",empty,"C"];

//Replace items by index
letters.splice(1, 2, "E", "I");// ["A","E","I"];

//empty array
letters.length = 0;
letters = [];
```

Array Looping

```
const numbers = [1, 4, 9, 16];
// for loop
for (let i = 0; i < numbers.length; i++) {
    console.log(numbers[i]);
}

const len = numbers.length;
for (let i = 0; i < len; i++) {
    console.log(numbers[i]);
}
// better is using foreach
numbers.forEach(num => console.log(num));
```

```
const doubles = numbers.map(function(num){
    return num * 2;
}); // return new Array [2, 8, 8, 32]

const odds = numbers.filter(function(item){
    return (item % 2) === 1;
}); // [ 16,4 ]

// sorting ascending
numbers.sort((a, b) => a - b);
// sorting decreasing
numbers.sort((a, b) => b - a);
```

Array Reduce

```
numbers = [2, 2, 3, 4];
const sum = numbers.reduce(function(total, num){
   return total + num;
});
// 11

let total = 0;
numbers.forEach(num => total += num);

let product = numbers.reduce(function (prev, current) {
    return prev * current;
});
// same as
product = numbers.reduce((prev, current) => prev * current);
```

Array common methods

```
let fruits = ['Apple', 'Banana'];
typeof fruits
// object

Array.isArray(fruits) // true

fruits.indexOf('Banana'); // 1
numbers.includes(4); // false

let clones = fruits.slice();
```

```
numbers = [1, 3, 5, 7, 9];
let squareNumber = numbers.find(function(num){
    return Number.isInteger(Math.sqrt(num));
});

let hasNegative = numbers.some(num => num < 0);

let areAllIntegers = numbers.every(num => num > 0);

let decreasingNumbers = numbers.reverse();
```

Array 2D

```
var matrix = [[]];
matrix["Toyota"] = ["Camry", "Corolla", "Prius"];
matrix["Honda"] = ["Civic", "Accord", "CR-V"];
matrix["Ford"] = ["F-150", "Mustang", "Escape"];

console.log(matrix["Toyota"][0]); // Camry
console.log(matrix["Honda"][2]); // CR-V
console.log(matrix["Ford"][1]); // Mustang
```

Object (reference type)

```
const person = {
    firstName: "John",
    lastName: "Doe",
    age: 50,
    eyeColor: "blue"
};
function hello(person){
    const { firstName, lastName } = person;
    console.log("Hi " + firstName + " " + lastName);
}
```

Contructor Pattern

```
function Player(userName, score) {
    this.userName = userName;
    this.score = score;
}

const kame = new Player('quylaokame', 1000);
kame.avatar = “./database/avatar/kame";

//iterate over an object
for (let prop in kame) {
  console.log(kame[prop]);
}
```

Object common methods

```
Object.create(source);
// Creates a new object extends source

Object.assign(target, source)
// Copies all values from source to target

Object.keys(obj)
// Returns an array containing all properties in obj

Object.values()
// Returns an array containing all values in obj

Object.freeze()
// Freezes an object.
```

Assignment

```
# Exercise 1

1. Write a function to format money string:
10000000 => “10,000,000" || 123456 => “123,456" || 12000.02 => "12,000.02"

2. Write a function for format money in shorten :
1000 => 1K, 1123400000 => 1.12B , 1342222 => 1.34M

3. Write the function to count how many words appear in a string:
oneTwoThree => 3

4. Write the function get the get the Extension of file:
“image.png” => “png”. “Sound.mp3” => “mp3”
```

Assignment

```
# Exercise 2

1. Write the function to calculate the combination (Ckn)

2. Write the function to get a random integer between 2 numbers: min, max;

3. Write the function get a random element from an arrays.

3. Given two arrays of integers,
find which elements in the second array are missing from the first array.
```

Assignment

```
# Exercise 3: 2D Arrays

- We have a rectangle garden with 3 row and 5 column:

- each cell had a bomb or no bomb: ( 0: SAFE, 1: BOMB)

- problem: find all the safe way to go from the left to the right of the
  garden.

- examples:
Input: [ [0,1,1], [0,1,1], [0,1,1], [0,1,1], [0,0,1]]
Output : [ [0,0,0,0,0] , [0,0,0,0,1]]
```

Assignment

```
# Exercise 4: Complete your test

1. convert to Roman number: n integer < 1,000

2. print how to read number in vietnamese: n integer < 1,000,000
726503 : bảy mươi hai vạn sáu ngàn năm trăm linh ba.
```

How to ask

```
1. What you want to do?

2. What did you do, and what is your issue?

3. Capture of your code or the error.
```